

Wenzheng WANG

PHD CANDIDATE IN ENGINEERING SCIENCES · EMBEDDED SYSTEMS ENGINEER

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Summary

PhD candidate at Sorbonne Université, holding an engineering degree (M.Sc.) in embedded systems and recipient of **two IEEE awards**. Specialized in mixed-signal electronics and signal processing for physiological signal sensing using optical spectroscopy. Developed a portable, wireless, multi-channel acquisition platform and a **real-time** analysis pipeline that addresses the fidelity-latency trade-off for continuous monitoring.

Education

PhD in Engineering Sciences

SORBONNE UNIVERSITY

- Design and Implementation of a Versatile Embedded Platform for Multimodal Physiological Signal Monitoring
- Advisors: Prof. S. Feruglio; Assoc. Prof. J. Denoulet

Paris, France

Nov. 2023 - Present

Master of Engineering in Embedded Systems

PARIS INSTITUTE OF DIGITAL TECHNOLOGY (ISEP)

- Graduated with honors (mention très bien)

Paris, France

2021 - 2023

Bachelor of Science in Vehicle Engineering

LANZHOU INSTITUTE OF TECHNOLOGY

- Ranked in the top 10% of the cohort (200 students)

Lanzhou, China

2016 - 2020

Professional Experience

PhD Researcher

SORBONNE UNIVERSITY COMPUTER SCIENCE LABORATORY (LIP6)

Development of an embedded physiological signal monitoring system on a dual-core platform, enabling 4 kHz multichannel sampling and sustained throughput of **416 kbps** with zero packet loss. Optimization of the wireless communication protocol by selecting a connection interval that is not an integer multiple of the acquisition and packetization periods, to mitigate electromagnetic interference, reducing artifacts by **54%** while preserving deterministic five-phase switching for the simultaneous acquisition of multi-wavelength PPG and ExG signals. Validation of the multi-wavelength approach on biological tissues via the **MOP-MCML** method. Development of an end-to-end Python analysis chain with a latency of **10 ms** (decoding, phase separation, artifact correction, quality assessment) and implementation of a **dual-path** estimation framework with quality-weighted fusion.

Paris, France

Nov. 2023 - Present

Teaching Assistant

SORBONNE UNIVERSITY & POLYTECH SORBONNE

Delivered **64 hours/year** across lectures, tutorials (TD), and labs (TP) for the BSc EEA and MSc SESI programs. Led Microcontrollers labs for 3 groups (**45 students**): prepared TP subjects, delivered hands-on sessions, and administered in-class quizzes. Ran Analog Electronics practicals for 4 groups (**60 students**), delivering 8-hour full-day TP per group and grading lab reports with quantitative feedback (32 hours/year). Taught Simulation of Electronic Circuits (14 hours/year) for 1 group (**13 students**), and proctored the final exam. Delivered an additional **guest lecture** to the final year MSc SESI cohort on emerging research topics.

Paris, France

Nov. 2023 - Present

Master's Internship

SORBONNE UNIVERSITY COMPUTER SCIENCE LABORATORY (LIP6)

Development of an **opto-electronic workflow** based on near-infrared optical spectroscopy to assess ankle ligament status after ligament reconstruction, coupling Monte Carlo modeling of light transport in biological tissues with experimental validation. Comparative evaluation of MCML and MCmatlab against analytical transmittance: multi-wavelength simulations, scanning source-detector spacing from 0.05 to 1.0 cm (0.05 cm step) to identify the optimal photon-path geometry across the range. Design of a motorized ankle positioning device using Fusion 360 and validation of its kinematics using a 3D-printed prototype.

Paris, France

Feb. 2023 - Sept. 2023

Bachelor's Internship

DONGFENG MOTOR CORPORATION

Supported casting and welding operations by tracking process parameters, assisting quality inspections, and maintaining documentation. Designed and revised vehicle driveshaft components in **CATIA/AutoCAD**, producing manufacturing-ready **3D drawings** with tolerancing.

Wuhan, China

Sept. 2019 - Dec. 2019

Skills

Languages

English (Professional working), French (Professional working), Chinese (Native)

Programming & Computing

C/C++, Python (scientific stack, signal processing + ML prototyping), R, Verilog/VHDL, CUDA

Embedded Platforms

Nordic Zephyr ecosystem, STM32 family, FPGA-based prototyping, dual-core architecture

Engineering Tools

Altium Designer, AutoCAD, Fusion 360, Keil MDK-ARM 5, LTspice, MATLAB/Simulink, Quartus, Solidworks

Publications

• JOURNAL

[1] A Review of Chronic Lateral Ankle Instability and Emerging Alternative Outcome Monitoring Tools in Patients following Ankle Ligament Reconstruction Surgery [PDF]

Journal of Clinical Medicine

I. SALIBA, A. HARDY, **W. WANG**, R. VIALLE, S. FERUGLIO

2024

This review analyzes clinical management of chronic lateral ankle instability and highlights limitations of current postoperative outcome assessment. It surveys emerging non-invasive monitoring modalities (e.g., bioelectrical/mechanical impedance and NIRS, a modality studied during my M2 internship) and outlines opportunities for more objective, longitudinal follow-up after ligament reconstruction. Work realized in collaboration with several hospital practitioners from AP-HP. DOI: 10.3390/jcm13020442

• INTERNATIONAL CONFERENCES

[2] Artifact-Robust SpO₂ Estimation from Multi-Wavelength Vertebral PPG: In Vivo Evaluation in a Porcine Model [PDF]

24th IEEE NEWCAS

W. WANG, N. MAINARD, J. DENOULET, M. BOUYER, M. GAUME, R. VIALLE, S. FERUGLIO

2026

Evaluated data processing pipelines for multi-wavelength transmissive vertebral PPG in a porcine model under hypoxia and tachycardia. Subspace filter achieved offline robustness (RMSE 3.21%), while legacy pipeline with auto-reset and 1D Kalman filtering provided optimal causal real-time SpO₂ estimation (RMSE 2.13%). DOI: 10.1109/NewCAS64543.2026.11674027

[3] Preliminary Development of an Opto-electronic System for Ankle Instability Assessment [PDF]

6th BioSMART

W. WANG, H. LI, I. SALIBA, A. HARDY, J. DENOULET, S. FERUGLIO

2025

Best Student Paper Award. Developed a preliminary opto-electronic approach for ankle instability assessment using NIRS modeling and simulation-driven design. The study provides evidence that optical signatures can support discrimination between ligament conditions, establishing a foundation for future wearable assessment tools. DOI: 10.1109/BioSMART66413.2025.11046076

[4] Design of a Multi-Channel Wireless System for Physiological Signals Acquisition [PDF]

23rd IEEE NEWCAS

W. WANG, H. LI, R. VIALLE, J. DENOULET, S. FERUGLIO

2025

First Place in Young Professionals Competition. Designed and implemented a multi-channel wireless platform for physiological signal acquisition, targeting high-fidelity sensing under constrained wireless latency and bandwidth. The system architecture and validation results support real-time mobile health monitoring and provide a scalable basis for edge-enabled analytics. No proceedings publication.

[5] Optimizing the Monte-Carlo simulation program for NIRS modeling of biological tissues in optoelectronic devices [PDF] [GitHub]

31st IEEE ICECS

W. WANG, S. LI, I. SALIBA, A. HARDY, R. VIALLE, J. DENOULET, S. FERUGLIO

2024

Optimized a Monte Carlo simulation workflow for NIRS modeling in biological tissues, improving usability and computational efficiency for opto-electronic design iterations. The updated tool was validated on single and multi-layer tissue scenarios, enabling more reliable optical propagation estimation for device prototyping. DOI: 10.1109/ICECS61496.2024.10849250

• NATIONAL CONFERENCES

[6] Système Opto-électronique Préliminaire pour l'Évaluation de l'Instabilité de la Cheville (in French) [PDF]

GDR SOC2

W. WANG, H. LI, I. SALIBA, A. HARDY, J. DENOULET, S. FERUGLIO

2025

Preliminary NIRS-based methodology and multi-layer tissue model for non-invasive ankle ligament assessment.

[7] Outil pour la conception de systèmes NIRS dans les tissus biologiques (in French) [PDF]

GDR SoC2

W. WANG, S. LI, I. SALIBA, J. DENOULET, S. FERUGLIO

2024

Design-oriented simulation toolchain for modeling light transport in biological tissues for NIRS optoelectronic systems.

Honors & Awards

Best Student Paper Award

Paris, France

6TH IEEE INTERNATIONAL CONFERENCE ON BIO-ENGINEERING FOR SMART TECHNOLOGIES (IEEE BioSMART)

2025

First Place in Young Professional Competition

Paris, France

23RD IEEE INTERNATIONAL NEW CIRCUITS AND SYSTEMS CONFERENCE (IEEE NEWCAS)

2025

Extracurricular Volunteer Experience

23rd IEEE International NEWCAS Conference

Paris, France

CONFERENCE STAFF, CO-RESPONSIBLE FOR VENUE SETUP, ATTENDEE COMMUNICATION AND REGISTRATION

2025

Paris 2024 Olympic and Paralympic Games

Paris, France

DRIVER, MEMBER OF THE PASSENGER FLEET DURING THE GAMES

2024

2021 MIDI Music Festival - Taihu Edition

Suzhou, China

STAGE CONTROL, MANAGED AUDIO SYSTEMS AND COORDINATED COMMUNICATIONS DURING PERFORMANCES

2021